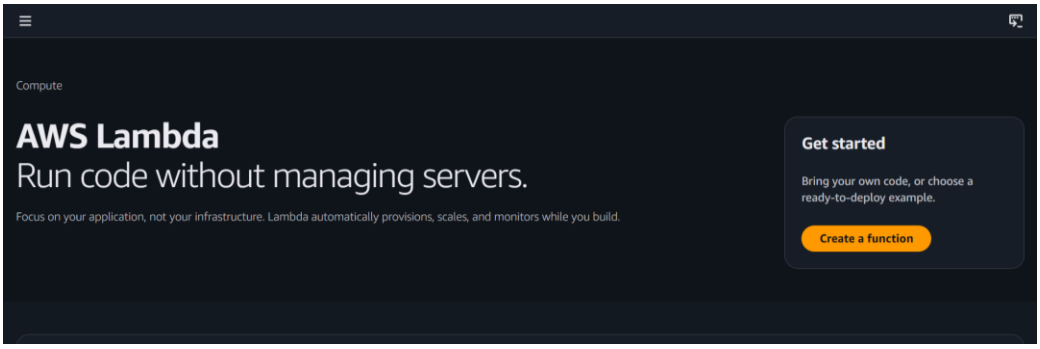
AWS Assignment Case Study 4

Canary Deployment Using Lambda and API Gateway

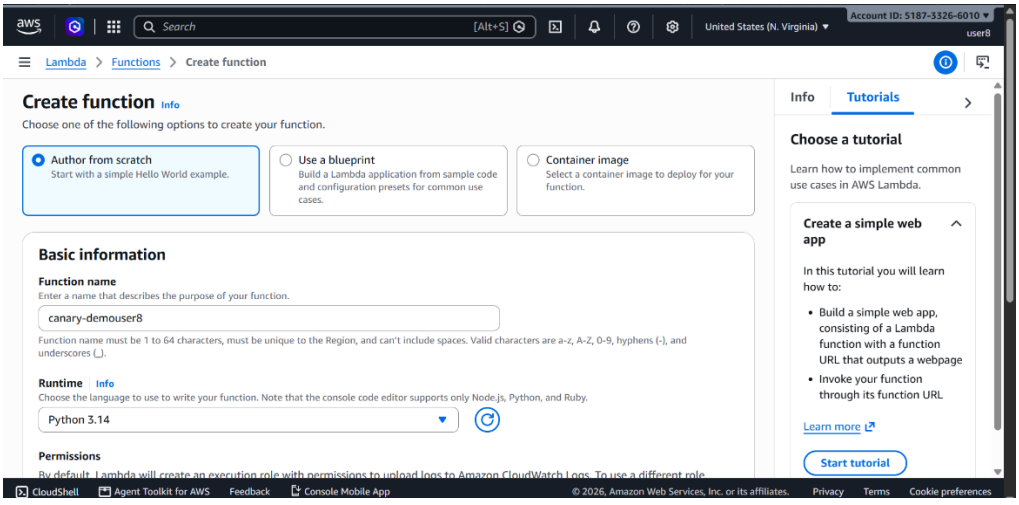
Problem Statement : The ABC Company is continuously updating the application software. The application is deployed on Lambda and API Gateway. Recently new version of the application is being released. The CTO of the company wants to check the new version of the application with a sample set of traffic and if everything is working fine then all 100% of traffic will be redirected. As application developer you are instructed to use native feature of API Gateway for doing so.

TASK 1: Create a Lambda function

1. Go to Lambda Console
2. Click on create function

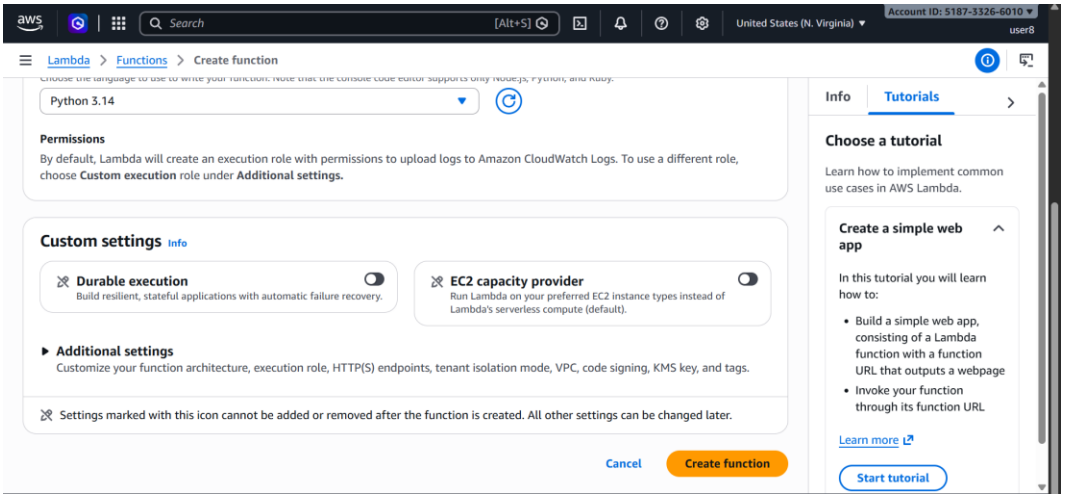


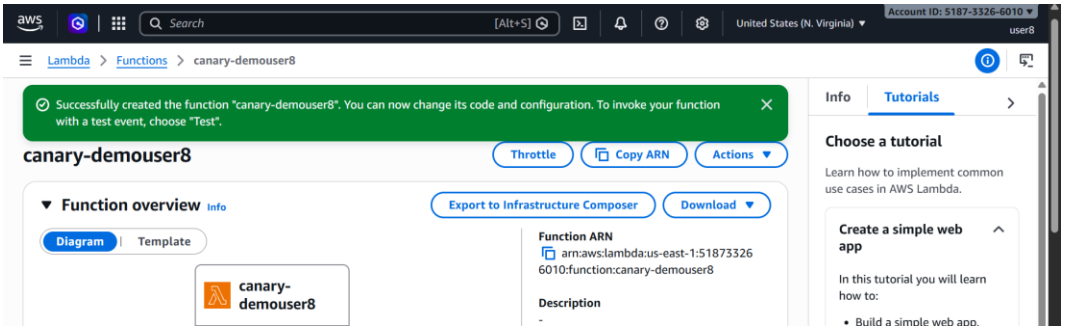
3. Choose author-from-scratch and your desired function name and select Python 3.10 or latest for Runtime.



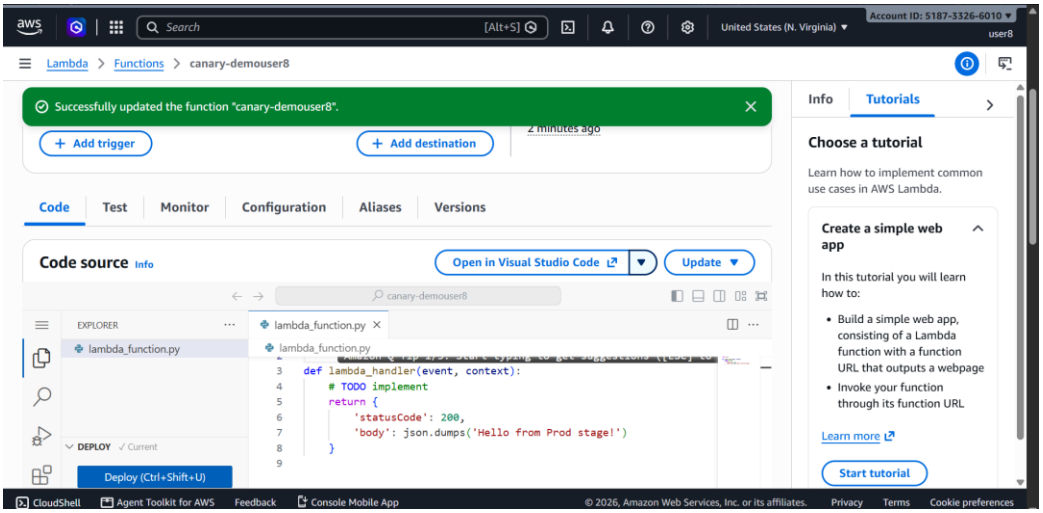
4. Click on Create function.

5.

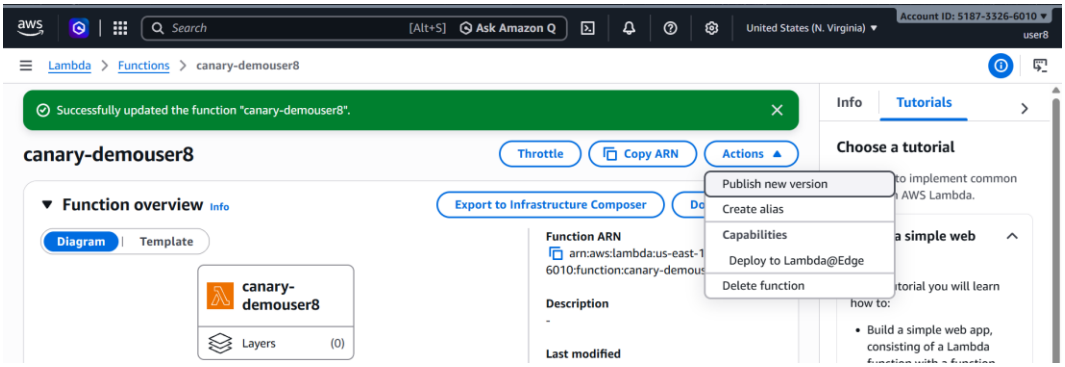


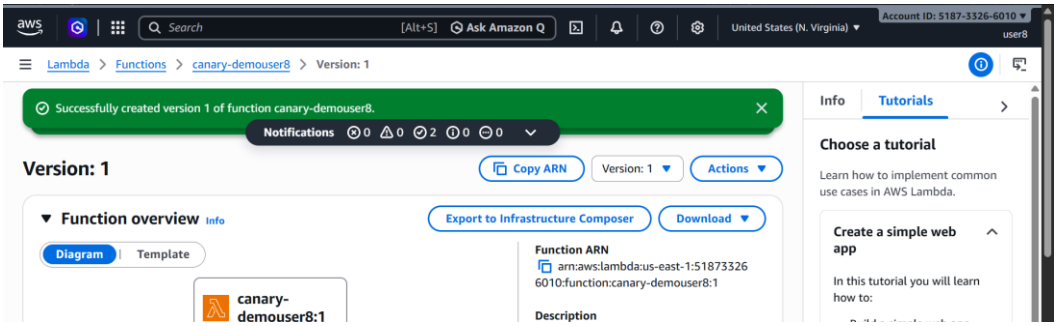


6. After the function is created, change the return to “Hello from Prod stage” instead of “Hello from Lambda”.

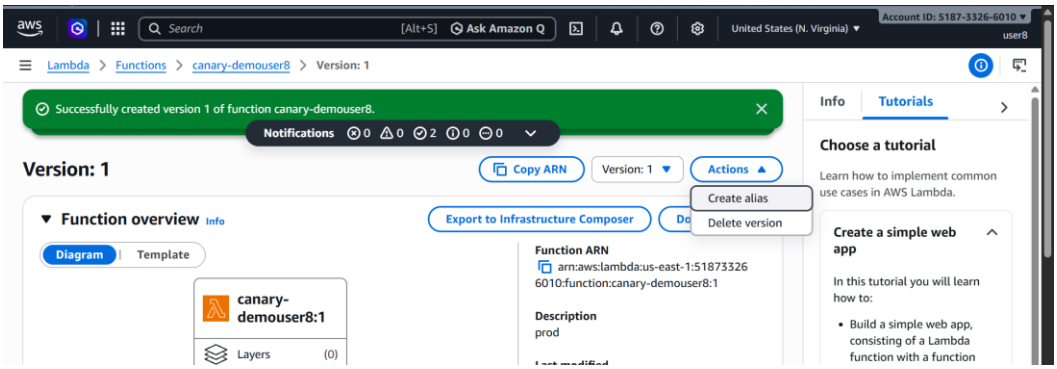


7. After successfully updating the function, from Actions select Publish a new version, put description as “Prod” and publish.

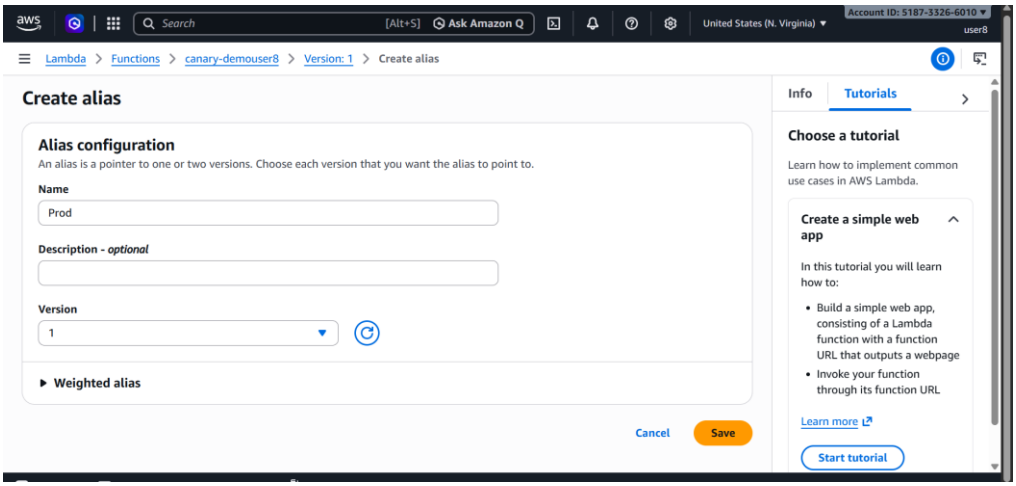


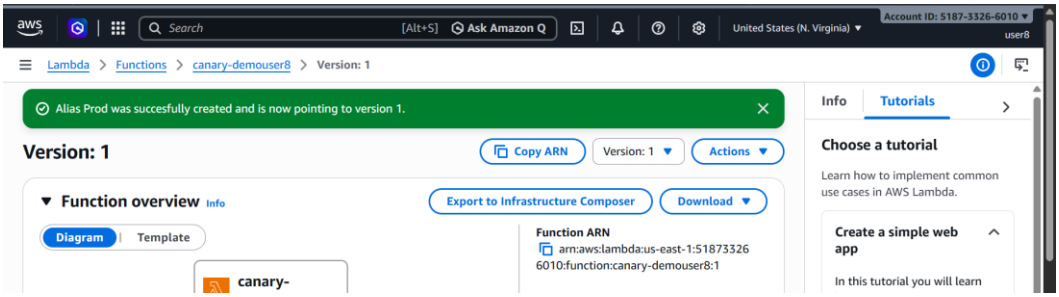


8. Go to action again and create an alias



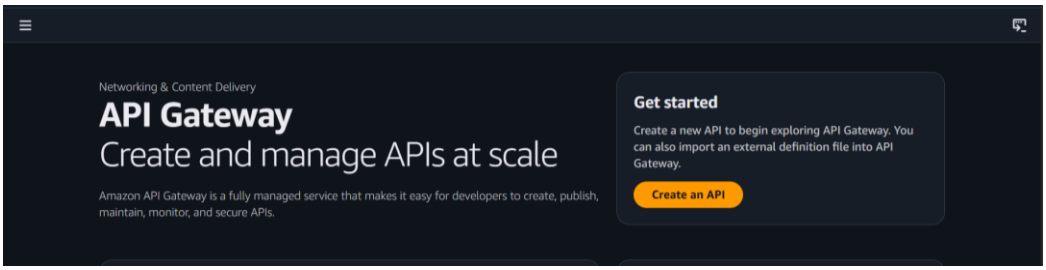
9. Enter name of alias as Prod, select Version 1 and save.



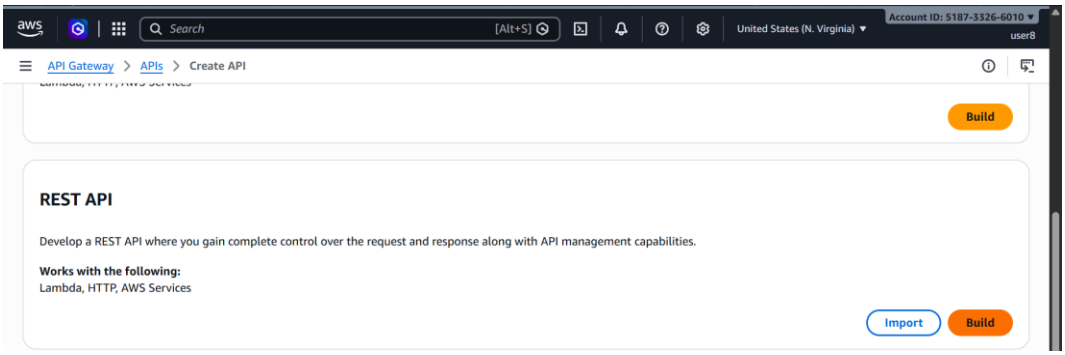


TASK 2: Create an API gateway

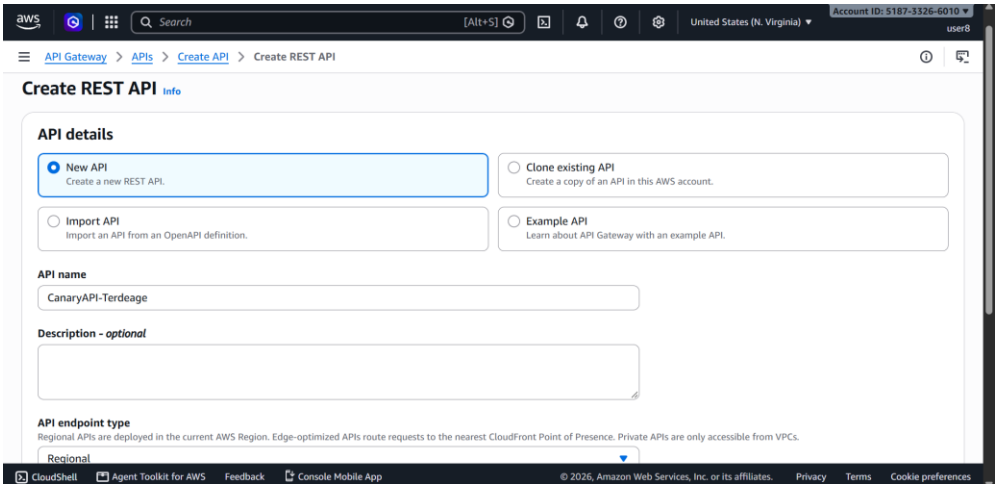
1. Go to API Gateway console and click on Create an API

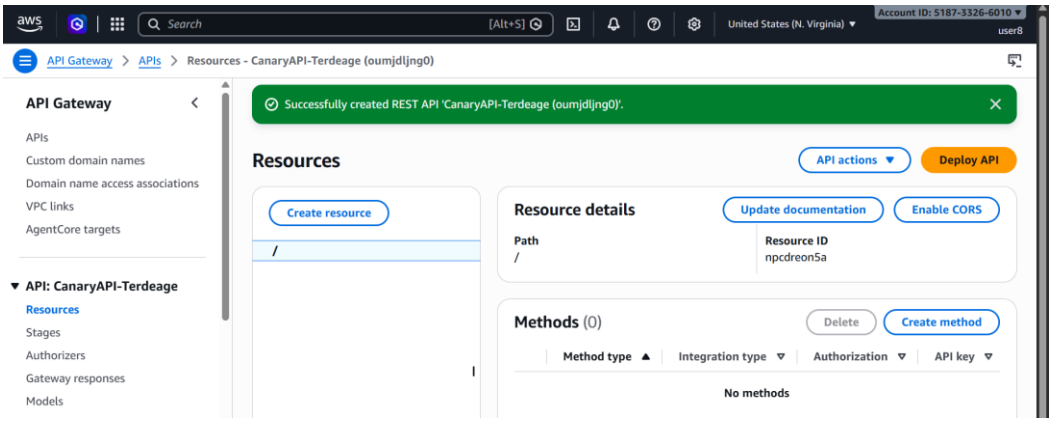


2. Select Rest API from the options and click on build.

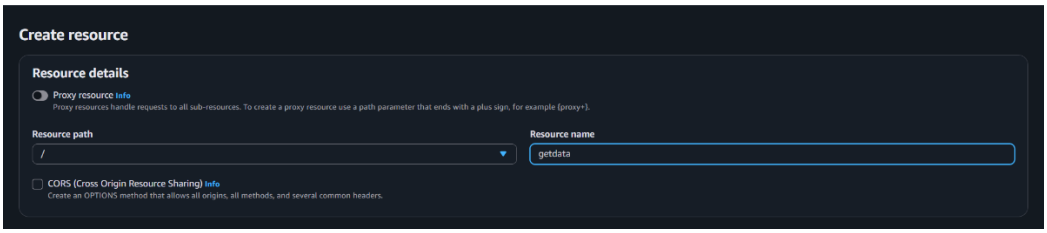


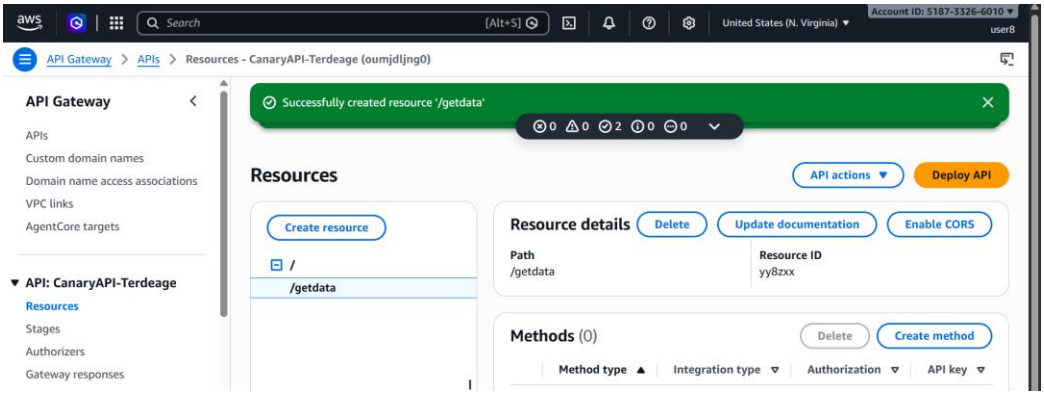
3. Select new API and name as whatever desired.



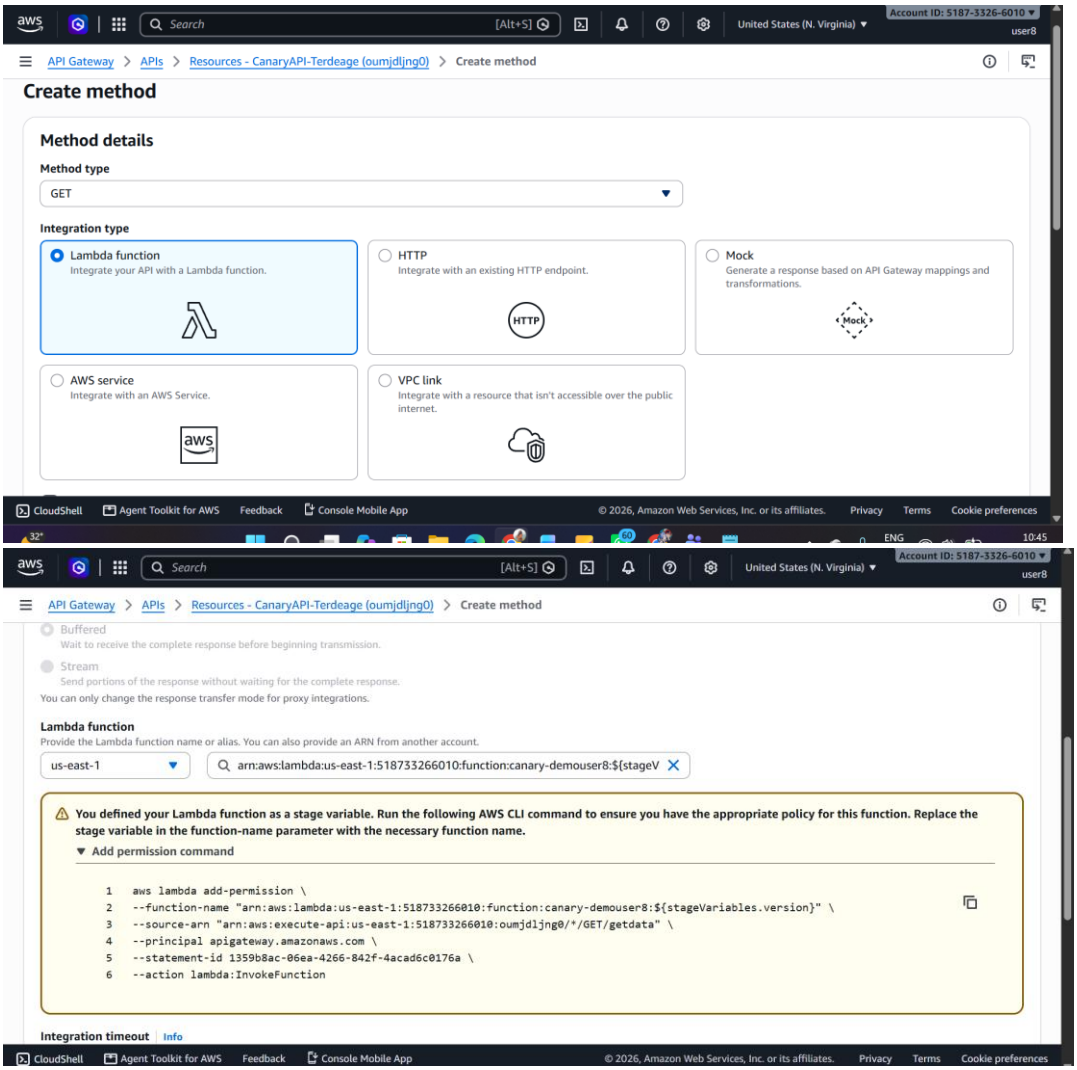


4. Click on Create Resource and give resource name as getdata



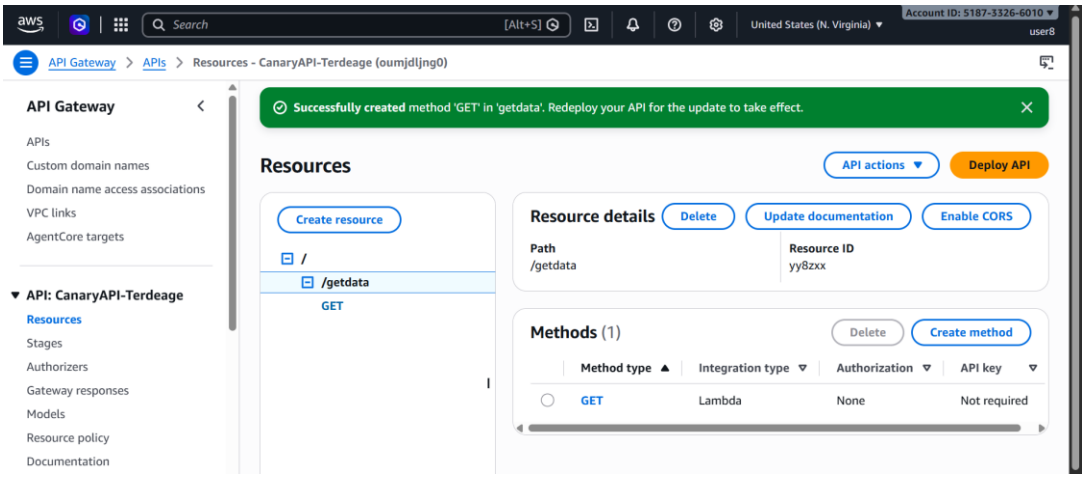


5. Now click on create method, select Method Type as GET, Integration type as Lambda and select the lambda which we have created from the drop down and add ":${stageVariables.version}" at the last where version is name of the stage variable and save.

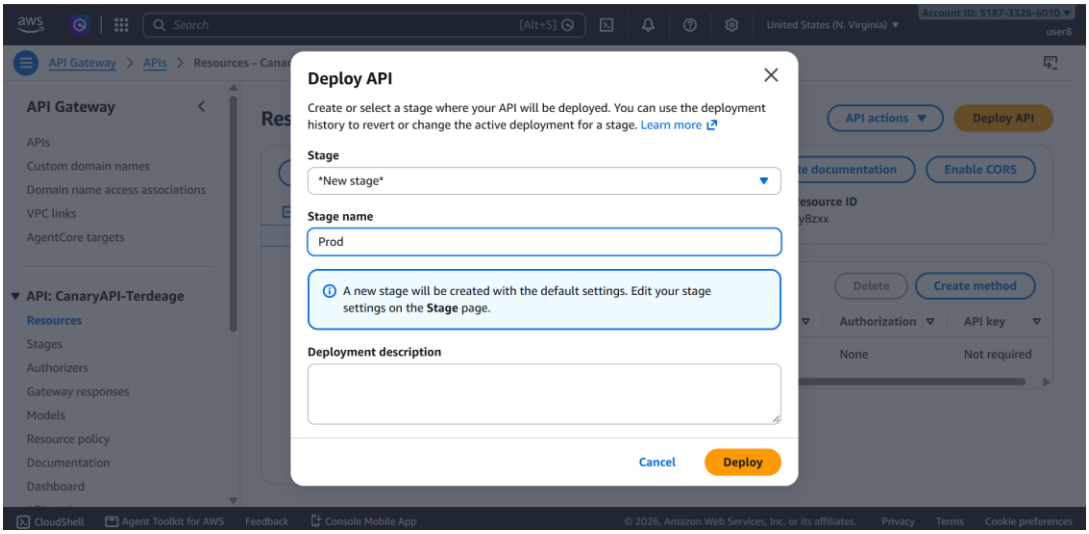


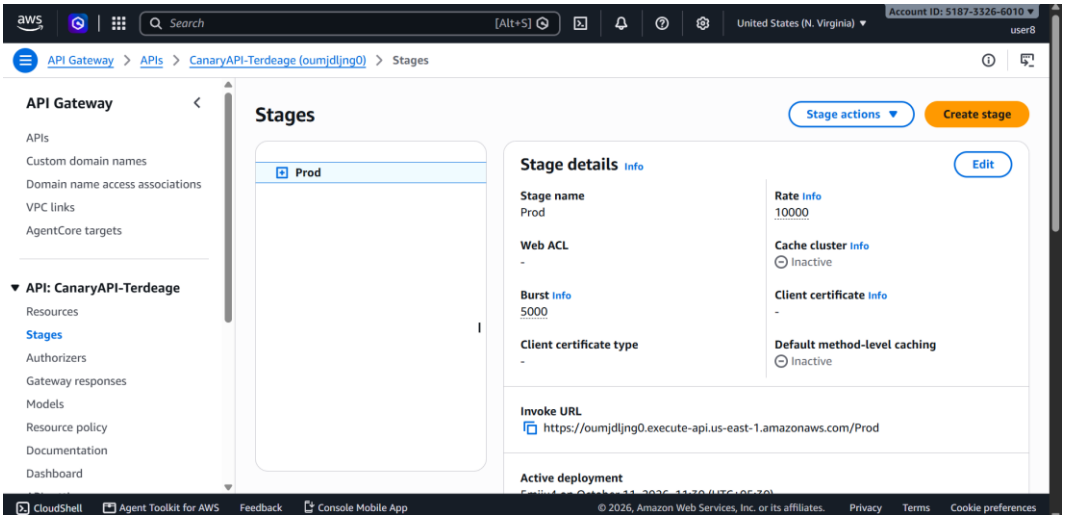
6. aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:canary-demouser8:Prod" --source-arn "arn:aws:execute-api:us-east-1:518733266010:oumjdljng0/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id 1359b8ac-06ea-4266-842f-4acad6c0176a --action lambda:InvokeFunction
   In place of stage version write the name of your Lambda version and run this in cloudshell



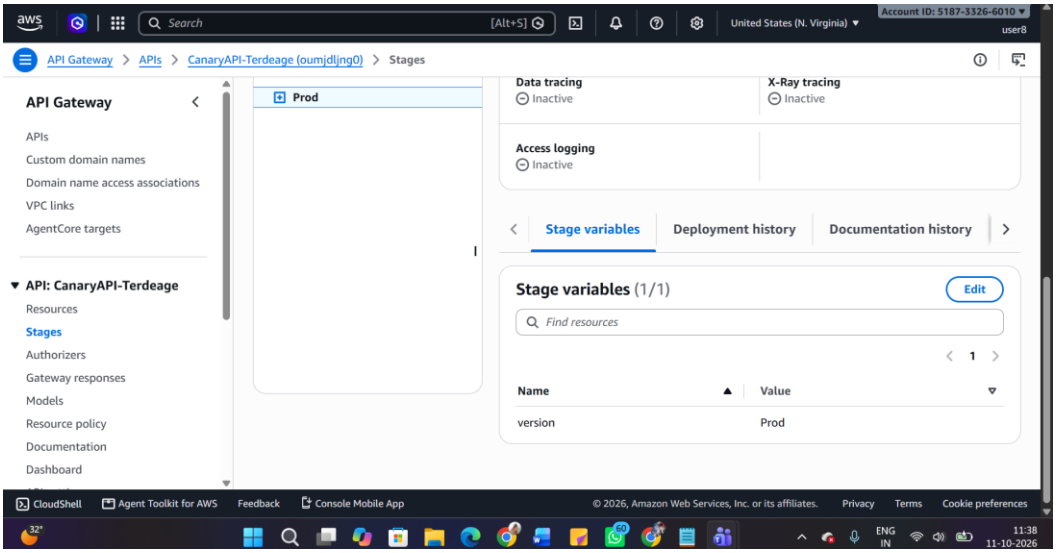


7. Now select Deploy API and select as New stage and give name as Prod and click on deploy.

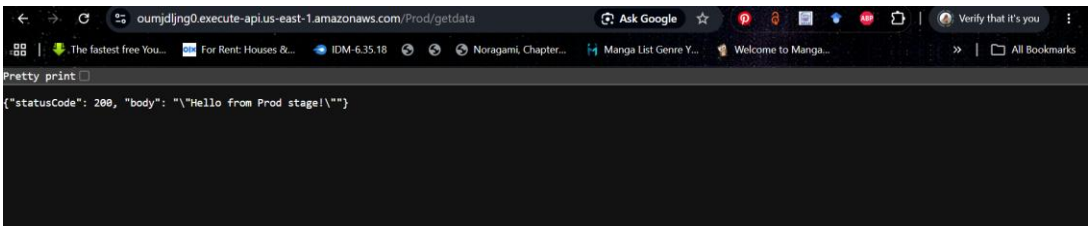




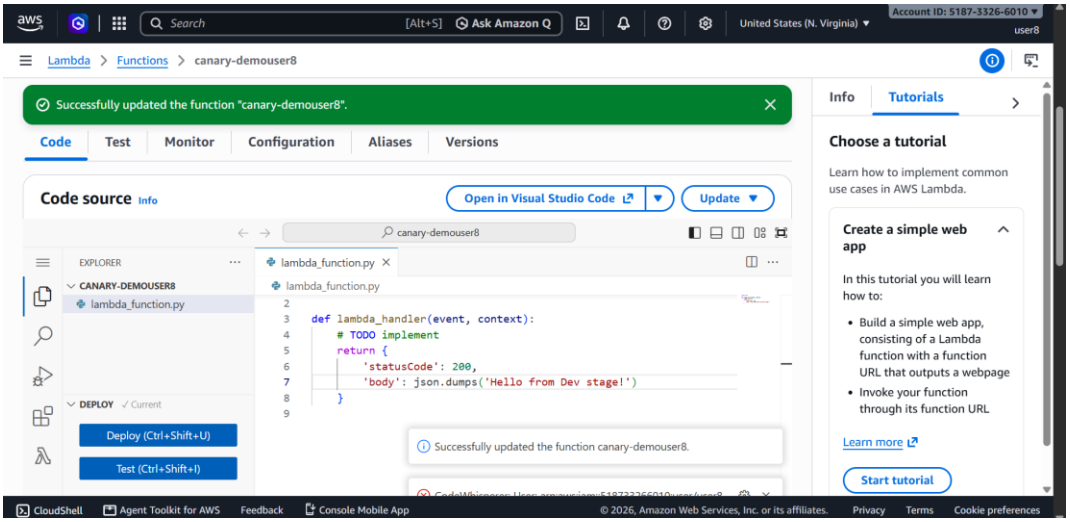
8. Next to add Stage variables click on Edit , then Add Stage Variable , give name as version and Value as Prod. Click on Save.



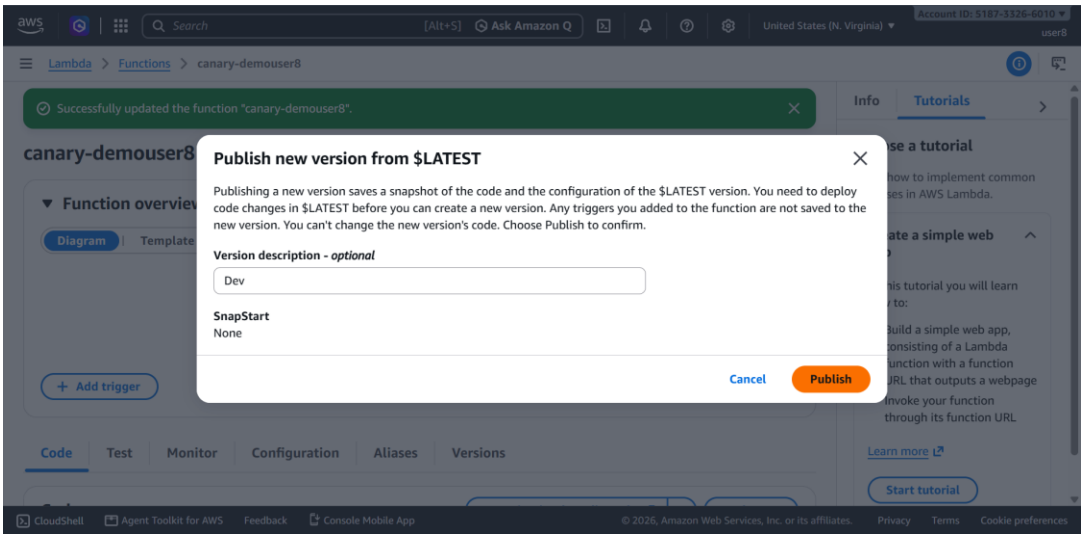
9. Now open the invoke URL in a new tab and at the end add getdata like this: https://oumjdljng0.execute-api.us-east-1.amazonaws.com/Prod/getdata
The output is as follows:

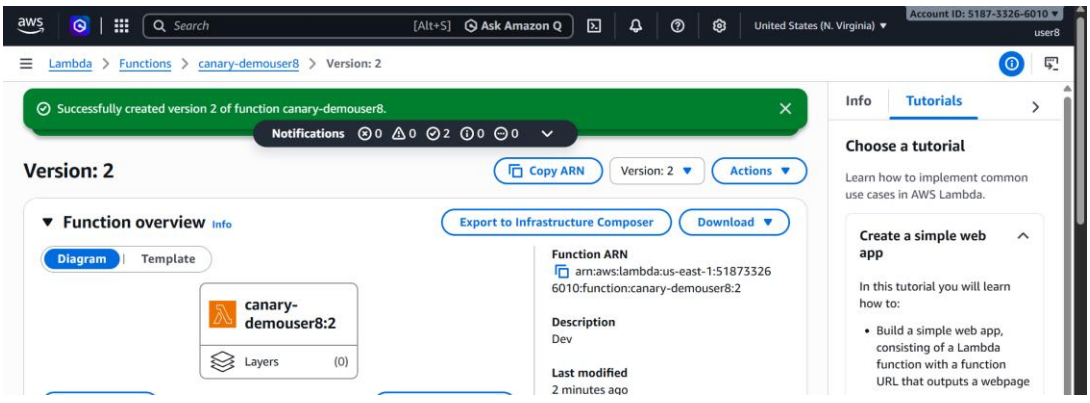


10. Now go the Lambda console and modify the latest code version.

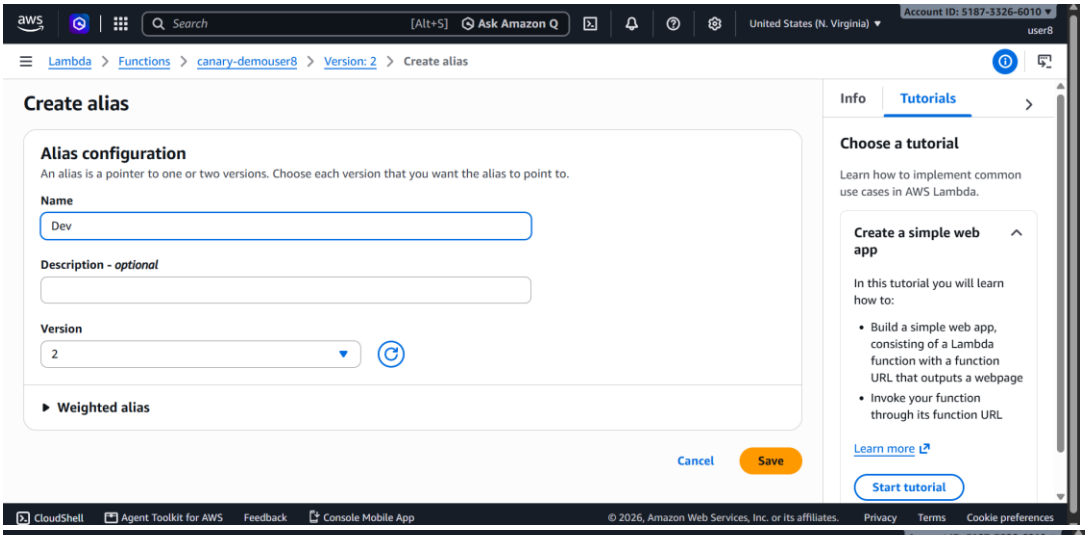


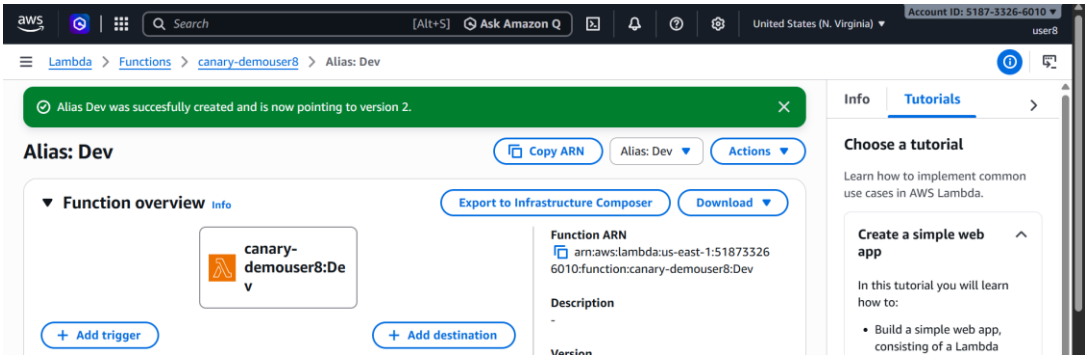
11. Go to actions again, select Publish a new version, Give description as Dev and click on Publish.



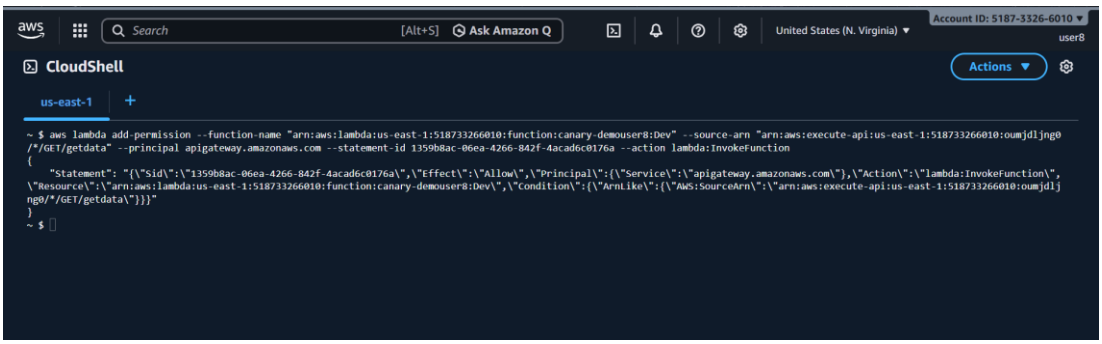


12. Go to action again, and Create alias. Under Create Alias, type name as Dev and Version as 2 and click on Save.

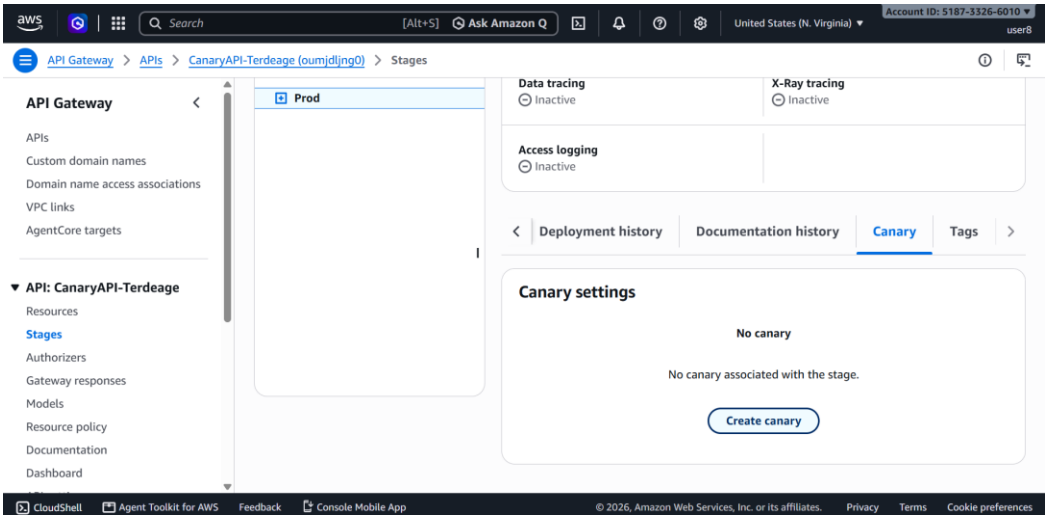




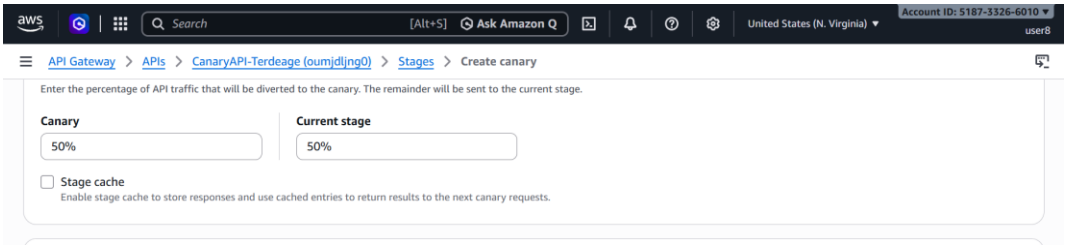
13. Again we need to add permission using AWS CLI
Instead of Prod replace it with Dev and run it.
aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:canary-demouser8:Dev" --source-arn "arn:aws:execute-api:us-east-1:518733266010:oumjdljng0//GET/getdata" --principal apigateway.amazonaws.com --statement-id 1359b8ac-06ea-4266-842f-4acad6c0176a --action lambda:InvokeFunction*



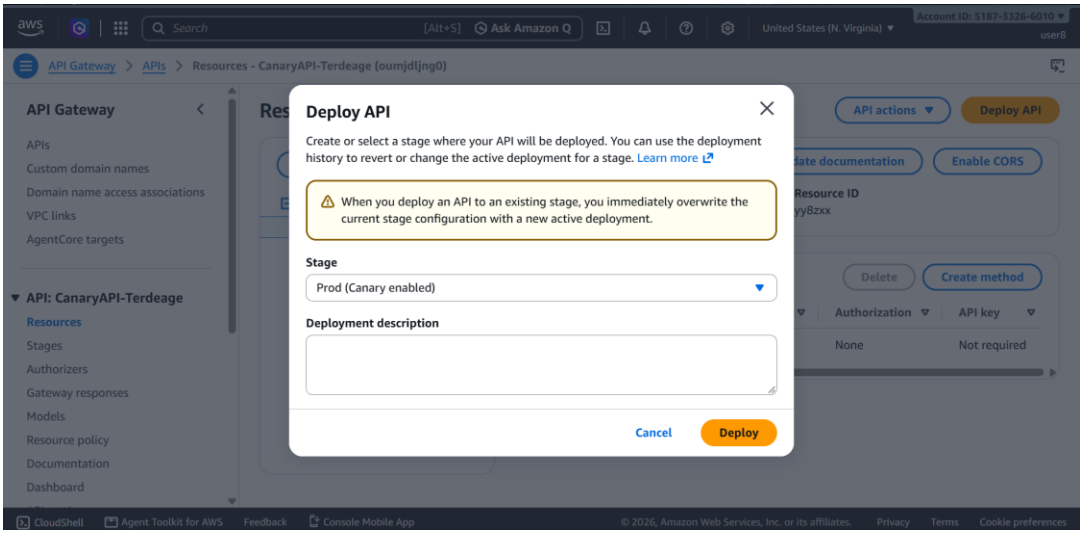
14. Now go to Canary tab in Prod stages and click on Create Canary.



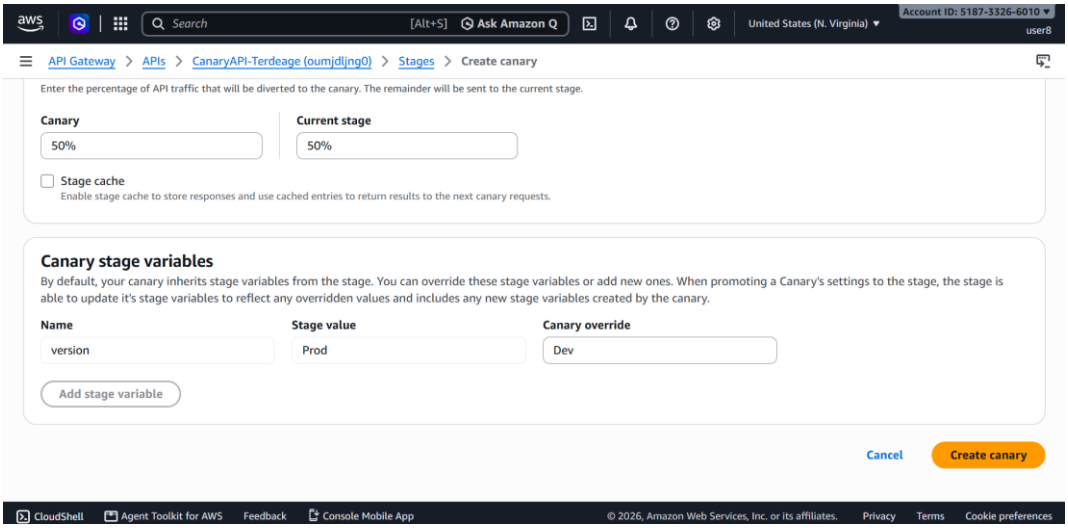
15. Now give distribution as 50% and click on Create canary.



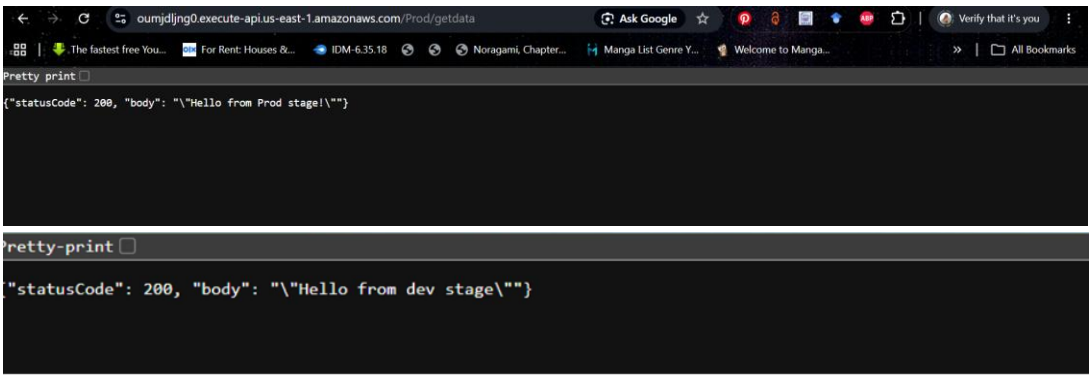
16. Now deploy the API again.



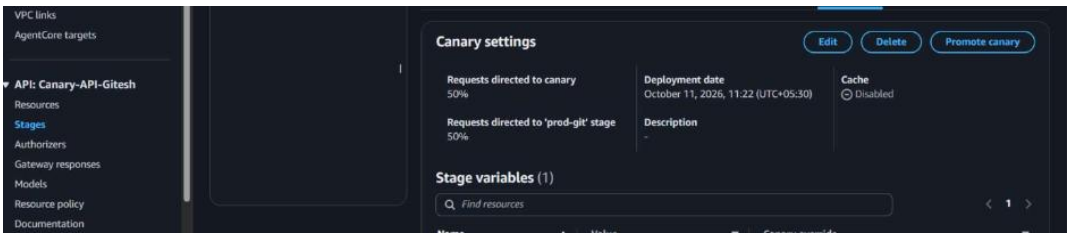
17. Now under the canary tab, click on Edit and write Canary override as Dev and click on Create Canary.



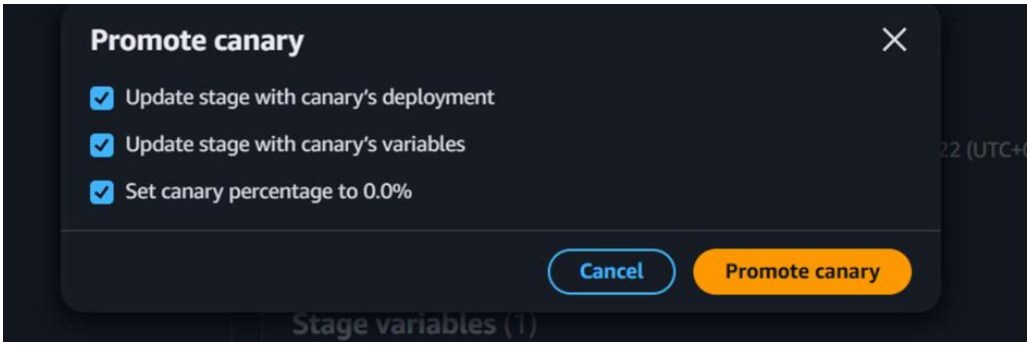
18. Now invoke the URL again appending /getdata at end, we will get “Hello from Dev stage” and “ Hello from Prod stage” randomly.



19. Now under the Canary, click on Promote Canary.



20. Select all options for update and Promote Canary.



21. Once you promote, you will notice under canary stage variables, both Stage value and Canary override value is Dev.



22. Now if you hit the URL again, we will get "Hello from Dev stage" every time.

